

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Part-III Examination, 2014-15

**Paper No. Paper-VIII: Numerical Analysis and Computer Programming
(Old Course)**

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

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- (a) Derive the Newton's forward difference interval formula for equal intervals. TAM see & fd Sea S as saad wa at efter asia Examination, 2014-15 . Statistics Unit - II Sang -T Old Course Paper : VIII 4. Derive the general quadrature formula and hence obtain Simpson's one third rule from it. frat vep-foes faq ast ort aise Numerical Methods 5. Establish Newton-Cotes formula for numerical receipt of this question paper) integration and herce obtain Trapezoidal rule A Note: Answer five questions in all. Question.No. from it. 1 is compulsory. Answer ONE question from aif Carat & fra Best Hea Gs Bl waa each unit. All questions carry equal marks. asa daw saa sitsigssra Faq at ora apis Beat wa sett & sere dis we Ge [#1] ama 2&1 7a sae A UH WS BH GAIT Unit - ITI . i : Be aA ust & sip sar [#1] gpg - Ill 6. Sum the following series to n terms :
- (b) ve oa and extrapolation. aan AP Bon va ar ast sa wT: SF at a afedsr at orearer difere |
- (c) Find the relationship between a and D

(a) 5, 10, 17, 28, 47, 82,, operator. aD ort & de aera ore aha BT/495

(b) Define factorial polynomial. (d (e (g (h

(c)) my Ua gese at oars diss | What is inverse interpolation ?Qt aa wT Ss ?Write the expression for the difference of product of two functions. a ort S WM H seat BH fra Shorey ferry | Evaluate 393 BT AT ATT HUT | Evaluate 2ab When interval of differencing is 1. SIT DI SHS AMAHLC 2gh HI AI aT ase | Find the general term of the series. Pet seh pr age ve att Hise

$2 + 6.4 + 12.8 + 20.16 + 30.32 + \dots$

Solve the following difference equation. Fa oetetinct a ea aia Hoa - Lux =0; $x > 0$ x

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- (a) Distinguish between the differential equations and difference equations. apa wich gq acdtipeit & dra Ae aaa | Unit - I go - I

- (a) If interval of differencing is one, prove that $\Delta^n f(x) = n! \Delta^n f(0)$ if $f(x)$ is a polynomial of degree n .
 (b) If $f(x)$ is a polynomial of degree n , prove that $\Delta^n f(x) = n! \Delta^n f(0)$.

- (a) Show that divided differences are symmetric functions of their arguments.
 (b) Describe Picard's method for solving ordinary differential equations. Obtain the expression for remainder term in Lagrange's interpolation formula.

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- (a)
 (b) BT/495

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[#1]

25.8 5.8.11 8.11.14 Solve the following difference equations.
 (i) $y_{n+1} - y_n = 2^n$

- (ii) (iii) 2

[#1]

Unit - IV

- (a) Obtain a cube root of 10 using Newton Raphson method.
 (b) Discuss Gauss-Jordan method for solving simultaneous linear equations.